MIL-B-2678B
29 APRIL 1966
SUPERSEDING
MIL-B-2678A
10 MARCH 1960
(SEE 6.5)

MILITARY SPECIFICATION

BRUSHES, WIRE, BOILER TUBE, (EXPANDING TYPE FOR USE WITH CLEANING-OUTFITS, BOILER-TUBE, ELECTRIC AND AIR MOTOR-DRIVEN)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers boiler tube wire expanding-type brushes for use with boiler tube-cleaning outfits.

1.2 Classification. Brushes shall be of the following classes, as specified (see 6.2):

Class A—For use with electric-motor-driven boiler tube cleaning-outfits.

Class B—For use with air-motor-driven boiler tube cleaning-outfits.

Class C—For use with either electric or air-motor-driven boiler tube cleaning outfits (with Universal joint).

Class D—For use with either electric or air-motor-driven boiler cleaning outfits.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-2684 —Tube and Flue Cleaner, Boiler, Internal, Electric Motor Driven.

MIL-H-15424—Hand Tools; Packaging of.

MIL-T-17870 —Tube and Flue Cleaner, Boiler, Internal, Air-Motor-Driven.

STANDARDS

FED-STD-151—Metals; Test Methods.

MILITARY

MIL-STD-105 —Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 —Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. — The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28—Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington D.C. 20402.)

FSC 5180

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3. REQUIREMENTS

3.1 Qualification for classes B, C and D brushes only.—The brushes furnished for classes B, C and D under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (4.2 and 6.3).

3.2 Preproduction sample for class A brushes only.—Prior to beginning production a complete brush sample shall be furnished for preproduction inspection as specified in 4.2 (see 6.4) for each class A size brush in the contract or order. The preproduction sample shall be identical to the proposed production items and shall be made and assembled by the same methods that are to be used on production of the item to fulfill the terms of the contract. Unless otherwise specified this sample shall be furnished for inspection and testing by a Government inspector in conjunction with the contractor at the contractor's plant. Should no facilities for conducting all of the tests at the contractor's plant be available, the sample shall be forwarded to the U.S. Naval Shipyard, Philadelphia, Pennsylvania (Naval Boiler and Turbine Laboratory) for that portion of the testing. The approved sample and the preproduction report shall be used as the production standard. Approval of the preproduction standard shall not relieve the contractor of his obligation to supply equipment conforming to this specification.

3.3 Material.—Materials to be used on brushes shall be as specified hereinafter.

3.4 Brushes shall be of an expanding and contracting type, providing automatic adjustment of the outside diameter of brushes to variations in the inside diameter of straight and curved sections of boiler tubes.

3.5 Dimensions.—Dimensions of brushes and the degree of expansion and contraction provided shall be such that, when operated with the corresponding type and size of an approved make of boiler-tube cleaning outfit, the brushes shall be capable of passing through and effectively operating in straight and curved sections of sizes of boiler tubes shown in table I (see 6.2).

TABLE I. *Dimensions of curved sections.*

Size of brush and outside diameter of boiler tube	Radius	Arc	Maximum crush of tube
Inches	*Inches*	*Degrees*	*Inch*
1	5	75	*0.050*
1⅛	5	75	.050
1¼	5	75	.062
1⅜	6	75	.062
1½	6	75	.075
1⅝	6	75	.075
1¾	6	75	.100
2	6	75	.120
2½	12	75	.140
2¾	12	75	.165
3 to 4	12	75	.190
4½	14	75	.215
5	16	75	.240

3.6 Brushes shall be efficient in removing deposits on the interior of boiler tubes with a minimum of damage to interior walls of tubes when operating in contact with bare metal walls.

3.7 All parts of brushes shall be made of steel. Working parts exposed to wear shall be of sufficient hardness and toughness to withstand the service required.

3.8 Bodies of brushes shall conform to the dimensions shown on figure 1 in order that refills may be interchangeable regardless of the manufacturer.

3.9 Refills. Refills shall be supplied in sets, each of which shall be composed of all the parts necessary to renew one complete brush (body, end caps, and screw not to be included).

3.10 Brushes shall be of such construction that the bristle sections can be renewed without renewing the brush body.

3.11 Bristles.—

3.11.1 Only wire bristles shall be used in brushes as the means of removing deposits on the interior of boiler tubes.

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TABLE II. *Chemical requirements.*

Carbon	Manganese	Phosphorus (max.)	Sulfur (max.)	Silicon
Percent	*Percent*	*Percent*	*Percent*	*Percent*
0.65 to 0.75	0.50 to 0.80	0.04	0.050	0.12 to 0.30

3.11.2 Bristles shall be of bright, tinned, high-tempered wire and shall conform to the chemical requirements shown in table II.

3.11.3 The wire shall conform to the following tensile strength and elongation:

Tensile strength — 250,000 to 300,000 p.s.i.
Elongation — 2 percent in 10 inches (min.)

3.11.4 Bristles shall be 0.0085 to 0.0095 inch in diameter in brushes of sizes up to and including 2 inches, and 0.016 to 0.020 inch in diameter in brushes of all larger sizes.

3.11.5 Ends of bristles shall not be trimmed by grinding.

3.12 Threaded connections.

3.12.1 Bodies of brushes shall be threaded for connection with flexible brush holders conforming to the requirements shown in table III.

3.12.2 *Threads.* — Dimensions of all threads, tolerances and gages for determining interchangeability shall comply with Handbook H-28 and conform to the requirements shown in table IV. Unified National Coarse series 2A-2B shall be the minimum or "loosest" acceptable standard for screw threading, UNC series 3A-3B or N8-3 the maximum or "tightest" standard for fit.

3.13 Drawings.—Master drawings covering assembly and detail of each brush shall be furnished.

3.14 Classes A, B and C (sizes up to and including 2 inches).

3.14.1 *Class A — for use with electric motor driven outfits only.*

3.14.1.1 Brushes of sizes up to and including 2 inches for use with electric-motor-driven outfits (Specification MIL-T-2684) shall conform to the dimensions shown in table V. Each brush shall be provided with

TABLE III. *Threading of bodies of brushes for connection with flexible brush holders.*

Class of brush	Size of brush	Size of thread	Threads per inch	Length of threaded section	Male or female
	Inches	*Inch*		*Inch*	
A, B and C	1 to 1½	5/16	18	5/16	Male
A, B and C	1⅝ to 1¾	7/16	14	5/8	Male
A	2	9/16	12	½	Female
B and C	2	7/16	14	5/8	Male
D	2½ to 4	9/16	12	½	Female
D	4½ to 5	⅞	9	9/16	Female

TABLE IV. *Dimensions.*

Tread size	Major diameter		Pitch diameter		Minor diameter (max.)
	Maximum	Minimum	Maximum	Minimum	
	Inch	*Inch*	*Inch*	*Inch*	*Inch*
9/16 by 18	0.3125	0.3043	0.2764	0.2734	0.2443
7/16 by 14	.4375	.4277	.3911	.3875	.3499
9/16 by 12	.5625	.5513	.5084	.5044	.4603
⅞ by 9	.8750	.8610	.8028	.7979	.7387

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a satisfactory mechanical means, independent of and in addition to the centrifugal force due to rotation of the brush for the automatic adjustment of the outside diameter of the brushes to variations in the inside diameter of the tubes.

3.14.2 *Class B — for use with air-motor-driven outfits only.*

3.14.2.1 Brushes of sizes up to and including 2 inches for use with air-motor-driven outfits (Specification MIL–T–17370) shall conform to the dimensions shown in table VI.

TABLE V. *Dimensions of class A brushes (sizes up to and including 2 inches).*

Outside diameter of tube	Wall thickness of tube	Inside diameter of tube	Diameter of brush		Free length of bristle (min.)	Wire ends per brush (min.)
			Expanded (min.)	Contracted (max.)		
Inches	*Inch*	*Inches*	*Inches*	*Inches*	*Inch*	*Number*
	0.085 to	0.810 to				
1	0.095	0.830	0.820	0.733	0.120	3360
1	0.109	0.782	.792	.705	.106	3360
1⅛	.109	.907	.917	.830	.169	3360
1⅛	.120	.885	.895	.808	.158	3360
1¼	.109	1.032	1.042	.920	.178	4800
1¼	.134	0.982	0.992	.870	.153	4800
1⅜	.120	1.135	1.145	1.022	.229	4800
1⅜	.134	1.107	1.117	0.994	.215	4800
1½	.120	1.260	1.270	1.110	.243	5760
1½	.134	1.232	1.242	1.082	.229	5760
1⅝	.134	1.357	1.367	1.172	.227	5760
1¾	.134	1.482	1.492	1.298	.290	5760
1¾	.156	1.438	1.448	1.254	.268	5760
2	.134	1.732	1.742	1.542	.287	8960
2	.165	1.670	1.680	1.480	.257	8960

TABLE VI. *Dimensions of class B brushes (sizes up to and including 2 inches).*

Outside diameter of tube	Wall thickness of tube	Inside diameter of tube	Diameter of brush		Free length of bristle (min.)	Wire ends per brush (min.)
			Expanded (min.)	Contracted (max.)		
Inches	*Inch*	*Inches*	*Inches*	*Inches*	*Inch*	*Number*
	0.085 to	0.810 to				
1	0.095	0.830	0.830	0.689	0.118	2560
1	0.109	0.782	.802	.661	.104	2560
1	.134	.732	.751	.610	.076	2560
1⅛	.109	.907	.927	.786	.167	2560
1⅛	.120	.885	.905	.764	.156	2560
1¼	.109	1.032	1.052	.894	.198	2560
1¼	.120	1.010	1.030	.872	.186	2560
1¼	.134	0.982	1.002	.844	.173	2560
1⅜	.120	1.135	1.155	.996	.249	2560
1⅜	.134	1.107	1.127	.968	.235	2560
1½	.120	1.260	1.280	1.120	.248	4800
1½	.134	1.232	1.252	1.092	.234	4800
1⅝	.134	1.357	1.377	1.176	.232	5760
1¾	.134	1.482	1.502	1.302	.295	5760
1¾	.156	1.438	1.458	1.258	.273	5760
2	.134	1.732	1.752	1.522	.292	8960
2	.148	1.704	1.724	1.494	.277	8960
2	.165	1.670	1.690	1.460	.262	8960
2	.180	1.640	1.660	1.430	.247	8960

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3.14.3 *Class C for use with either electric or air-motor-driven outfits.*

3.14.3.1 Class C brushes for use with either electric-motor-driven or air-motor-driven outfits shall conform to the dimensions shown in table VII. A universal joint of a type satisfactory to the Bureau or agency concerned shall be attached to the end of the brush that joins the brush holder, to permit satisfactory operation in tubes bent to a radius of 2¼ inches through 180 degrees when the maximum crush of tube does not exceed 50 percent more than that shown in table I.

3.15 Brush class D (above 2 inches in size).

3.15.1 *Class D for use with either electric or air-motor-driven boiler tube cleaning outfits.*—Brushes of sizes larger than 2 inches, up to and including 6 inches, for use with class D electric-motor-driven or air-motor-driven outfits, shall conform to the dimensions shown in table VIII.

TABLE VII. *Dimensions of class C brushes (sizes up to and including 2 inches).*

Outside diameter of tube	Wall thickness of tube	Inside diameter of tube	Diameter of brush		Free length of bristle (min.)	Wire ends per brush (min.)
			Expanded (min.)	Contracted (max.)		
Inches	*Inch*	*Inches*	*Inches*	*Inches*	*Inch*	*Number*
	0.035 to	0.810 to				
1	0.095	0.830	0.810	0.705	0.125	1920
1	0.109	0.782	.782	.677	.110	1920
1	.134	.732	.732	.627	.080	1920
1⅛	.109	.907	.907	.802	.178	1920
1⅛	.120	.885	.885	.780	.162	1920
1⅛	.109	1.032	1.042	.956	.163	1920
1¼	.134	0.982	0.992	.906	.138	1920
1⅜	.120	1.135	1.145	1.059	.214	1920
1⅜	.134	1.107	1.117	1.031	.200	1920
1½	.120	1.260	1.270	1.070	.220	3840
1½	.134	1.232	1.242	1.042	.206	3840
1⅝	.134	1.357	1.367	1.125	.205	3840
1¾	.134	1.482	1.492	1.246	.240	3840
1¾	.156	1.438	1.448	1.200	.220	3840
2	.134	1.732	1.742	1.422	.321	6400
2	.148	1.704	1.714	1.394	.305	6400
2	.165	1.670	1.680	1.360	.290	6400

3.16 Workmanship.—Workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.—Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification tests[1] (for class B, C and D brushes only).—Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. Qualification tests shall consist of the examination of 4.5.1 through 4.5.1.3 and the tests of 4.7.2 and 4.7.3. Tests

[1] Application for qualification tests shall be made in accordance with "PROVISIONS GOVERNING QUALIFICATION" (see 6.3 and 6.3.1).

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TABLE VIII. *Dimensions of class D brushes (sizes larger than 2 inches, up to and including 5 inches).*

Outside diameter of tube	Wall thickness of tube	Inside diameter of tube	Diameter of brush		Free length of bristle (min.)	Wire ends per brush (min.)
			Expanded (min.)	Contracted (max.)		
Inches	*Inch*	*Inches*	*Inches*	*Inches*	*Inch*	*Number*
2½	0.120	2.260	2.869	1.924	0.307	1800
2½	.134	2.232	2.841	1.896	.293	1800
2½	.148	2.204	2.813	1.868	.279	1800
2½	.165	2.170	2.279	1.834	.262	1800
2¾	.120	2.510	2.619	2.174	.432	1800
2¾	.134	2.482	2.591	2.146	.418	1800
2¾	.148	2.454	2.563	2.118	.404	1800
2¾	.165	2.420	2.529	2.084	.387	1800
3	.134	2.732	2.825	2.417	.391	2520
3	.165	2.670	2.763	2.355	.360	2520
3	.250	2.500	2.667	2.259	.312	2520
3¼	.134	2.982	3.075	2.667	.516	2520
3¼	.165	2.920	3.013	2.605	.485	2520
3¼	.250	2.750	2.843	2.435	.400	2520
3½	.120	3.260	3.353	2.902	.428	2520
3½	.134	3.232	3.325	2.874	.412	2520
3½	.165	3.170	3.263	2.812	.381	2520
3½	.180	3.140	3.233	2.782	.365	2520
4	.134	3.732	3.825	3.374	.662	2520
4	.165	3.670	3.763	3.312	.631	2520
4	.203	3.594	3.687	3.236	.593	2520
4	.250	3.500	3.593	3.142	.546	2520
4½	.180	4.140	4.233	3.729	.578	2520
4½	.203	4.094	4.187	3.683	.555	2520
4½	.220	4.060	4.153	3.649	.538	2520
4½	.250	4.000	4.093	3.589	.508	2520
4½	.280	3.940	4.033	3.529	.478	2520
4½	.293	3.914	4.007	3.503	.465	2520
4½	.343	3.814	3.907	3.403	.415	2520
5	.300	4.400	4.493	3.989	.415	2520

shall be conducted on each size and class of brush for which qualification approval is desired.

4.3 Preproduction inspection and testing of each size of class A brush only.—Each preproduction sample (see 3.2) shall be visually and dimensionally examined to verify conformance with 4.5 and tested to verify conformance with 4.7.1 and 4.7.3.

4.4 Sampling for quality conformance inspection.

4.4.1 *Lot.*—All brushes or refill sets of one class and size offered for delivery at one time shall be considered a lot for purposes of quality conformance inspection.

4.4.2 *Sampling for visual examination.*—A random sample of brushes or refill sets shall be selected from each lot in accordance with MIL–STD–105 at inspection level S–1 and Acceptable Quality Level (AQL) of 1.5 percent defective. Each of the sample units shall be subjected to the visual examination specified in 4.5.1.

4.4.3 *Sampling for dimensional examination.* A random sample of brushes or refill sets shall be selected from each lot in accordance with MIL–STD–105, at inspection level S–4 and Acceptable Quality Level of 2.5 percent defective. Each of the sample units shall be subjected to the dimensional examination specified in 4.5.1.1.

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FIGURE 1. *Boiler tube expanding type body.*

Dimensions of class A — Sizes up to and including 2 inches.

Brush size	A	B	C	D	Number of slots
Inches	*Inch*	*Inches*	*Inches*	*Inch*	
1 and 1 1/16	5/32	9/16	1 15/16 ± 0.005	0.159 + 0.003 − .000	6
1 1/8 and 1 5/16	5/32	11/16	1 11/16 ± .005	.222 + .003 − .000	6
1 1/8	5/32	13/16	1 13/32 ± .005	.222 + .003 − .000	6
1 5/16 and 1 3/8	1/8	15/16	1 27/64 ± .005	.224 + .003 − .000	6
2	5/16	1 5/16	1 5/8 ± .005	.224 + .003 − .000	8

Dimensions of class B (sizes up to and including 2 inches)

Brush size	A	B	C	D	Number of slots
Inches	*Inch*	*Inches*	*Inches*	*Inch*	
1 and 1 1/16	1/16	9/16	45/64 ± 0.005	0.222 + 0.003 − .000	4
1 1/8 and 1 5/16	5/64	11/16	45/64 ± .005	.222 + .003 − .000	4
1 1/8	1 1/64	13/16	1 1/32 ± .005	.222 + .003 − .000	6
1 5/16 and 1 3/8	3/16	15/16	1 27/64 ± .005	.224 + .003 − .000	6
2	3/8	1 5/16	1 5/8 ± .005	.224 + .003 − .000	8

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Dimensions of class C (sizes up to and including 2 inches)

Brush size	A	B	C	D	Number of slots
Inches	*Inch*	*Inches*	*Inches*	*Inch*	
1 and 1⅛	1/16	9/16	51/64 ± 0.005	0.222 + 0.003 − .000	4
1¼ and 1⅜	9/64	5/8	51/64 ± .005	.222 + .003 − .000	4
1½	5/16	¾	63/64 ± .005	.222 + .003 − .000	6
1⅝ and 1¾	¼	1	63/64 ± .005	.222 + .003 − .000	6
2	9/32	1⅛	1 1/64 ± .005	.222 + .003 − .000	8

Dimensions of class D (sizes larger than 2 inches, up to and including 5 inches)

Brush size	A	B	C	D	Number of slots
Inches	*Inch*	*Inches*	*Inches*	*Inch*	
2½	½	1 9/16	1 51/64 ± 0.005	0.442 + 0.005 − .000	6
3 and 3¼	5/8	2	2 3/32 ± .005	.570 + .005 − .000	6
3½ and 4	¾	2⅜	2 3/32 ± .005	.570 + .005 − .000	6
4 and 4½	1 9/16	3	2 ⅛ ± .005	.570 + .005 − .000	6
5	1 9/32	3⅜	2 ⅛ ± .005	.570 + .005 − .000	6

4.4.4 *Sampling for bristle count.*—A random sample of brushes or refill sets shall be selected from each lot in accordance with MIL–STD–105, at inspection level S–2 and shall be subjected to the examination specified in 4.5.1.2.

4.5 Inspection.

4.5.1 *Visual examination.*—Each of the sample brushes or refill sets selected in accordance with 4.4.2 shall be subjected to visual examination (other than dimensional examination and bristle count) to determine compliance with this specification. If the number of nonconforming brushes exceeds the acceptance number shown in 4.4.2, the entire lot shall be rejected.

4.5.1.1 *Dimensional examination.*—Each of the samples selected in accordance with 4.4.3 shall be subjected to a careful dimensional check to determine conformance with the applicable dimensional requirements of 3.8, 3.11.4 and tables III to VIII, inclusive.

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Note. In checking the free length and diameter of the bristle, this measurement may be made on only 1 bristle cartridge from each sample unless the inspector considers further examination desirable. If the number of nonconforming brushes or refill sets exceeds the acceptance number specified in 4.4.3 the entire lot shall be rejected.

4.5.1.2 *Bristle count.*—The number of wire ends, in one faggot of each of the sample brushes selected in accordance with 4.4.4, shall be carefully counted. The faggot to be counted shall be selected from a different row in each of the sample brushes or refill sets. The number of wire ends per brush shall then be calculated by multiplying the count of wire ends by number of faggots in the corresponding brush. If the resulting number is not in conformance with the requirements of tables V to VIII inclusive, the entire lot shall be rejected.

4.5.1.3 *Visual and dimensional examination and bristle count.*—Examination shall be in accordance with table IX.

TABLE IX. *Classification of defects in accordance with MIL-STD-105*

Categories	Defects	Visual	Dimensional	Bristle count	Sampling paragraph
Critical:					
1	None defined.				
Major:					
101	Evidence of use of unauthorized material	X			4.4.2
102	Brushes not class specified; not expanding and contracting type.	X			4.4.2
103	Brush body dimensions and configuration not in conformance with approved figure; not within tolerances specified.		X		4.4.3
104	Bristles not bright or tinned	X			4.4.2
105	Brush diameter not within tolerance specified		X		4.4.3
106	Evidence of brushes being trimmed by grinding	X			4.4.2
107	Brush diameters (expanded and contracted) free length of bristles and number of wire ends per brush nonconforming.		X	X	4.4.3 and 4.4.4
108	Bodies of brush not thread size, threads per inch specified, length of threaded section nonconforming.		X		4.4.3
109	Threads not of class or fit specified; major, minor and fitted diameters not within the tolerance specified.		X		4.4.3
110	Universal joint nonconforming (for classes A and B universal brushes) not attached as specified.	X			4.4.2
111	Bristle sections cannot be renewed without renewing body.	X			4.4.2
112	Refill sets not complete; units missing (if required)	X			4.4.2
113	Brushes do not comply with approved drawings	X	X		4.4.2 and 4.4.3
114	Workmanship poor	X			4.4.2
115	Automatic adjustment of outside of brushes not operatable.	X			4.4.2
116	Parts of refill sets not correct count	X			4.4.2
Minor:					
201	None defined.				

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4.6 Comparison testing.—When deemed necessary, one sample complete brush and wire shall be selected and forwarded to a laboratory satisfactory to the Bureau of Ships for comparison tests. The sample shall be subjected to any test considered necessary to determine that the sample is equal to the sample upon which qualification approval was based. If an unsatisfactory report is obtained, the Bureau of Ships shall be notified immediately.

4.7 Tests.

4.7.1 *Preproduction test for class A brushes only.*—Each preproduction sample (see 4.3) shall be fitted or assembled into a complete electric motor driven tube and flue cleaner suitable for the sample brush or refill set (cleaner to conform to MIL-T-2684). It shall then be visually demonstrated that the sample is an acceptable part of the total assembly when the tube cleaner is inserted in a minimum of one foot length of boiler tube of the size designated by the sample.

4.7.2 *Qualification test for class B, C and D brushes only.*—Each sample selected in accordance with 4.4.1 shall be assembled or fitted into a standard cleaning unit selected from stock suitable for brush or refill set size. The brush shall be capable of passing through and effectively operating in straight and curved sections of sizes of boiler tubes for which designated. This sample shall be operated in cleaning of dirty boiler tubes for 5 hours. Both the sample and the tubes involved shall then be inspected to determine the acceptability of the sample as compared to the condition and action of qualified standard stock brushes after similar operation for 5 hours.

4.7.3 *Wire.*—Suitable specimens of the bristle wire for brushes of each size group as specified in 3.11.4 shall be tested and analyzed to determine conformance with 3.11. Physical tests shall be conducted in accordance with the applicable requirements of FED-STD-151.

4.8 Examination of preparation for delivery.—An examination shall be made to determine that packaging, packing and marking are in compliance with section 5. Unless otherwise specified, the inspection level shall be level S-2 with an acceptable quality level (AQL) of 4.0 percent defective.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing.

5.1.1 Brushes and brush sections (refills) shall be preserved and packaged level A or C, as specified (see 6.1), packed level A, B or C as specified (see 6.2) in accordance with MIL-H-15424. Brushes shall be packaged by units, section refills by sets sufficient to renew one brush.

5.2 Marking.—In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use.—Brushes and brush sections covered by this specification are intended for use in the cleaning of the internal surfaces of boiler tubes, condenser tubes, heat exchanger tubes, and the like.

6.2 Ordering data.—Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Class required (see 1.2).

(c) Dimensions of curved sections (see 3.5).

(d) Dimensions of brushes (see tables V to VIII inclusive).

(e) Applicable levels of preservation, packaging and packing (see 5.1).

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in Qualified Products List QPL 2678. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to

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have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Chief of the Bureau of Ships, Department of the Navy, Washington, D.C. 20360, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.4 Preproduction.

6.4.1 Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.5 Changes from previous issue.—The extent of changes (deletions, additions, etc.) preclude the annotation of the individual changes from the previous issue of this document.

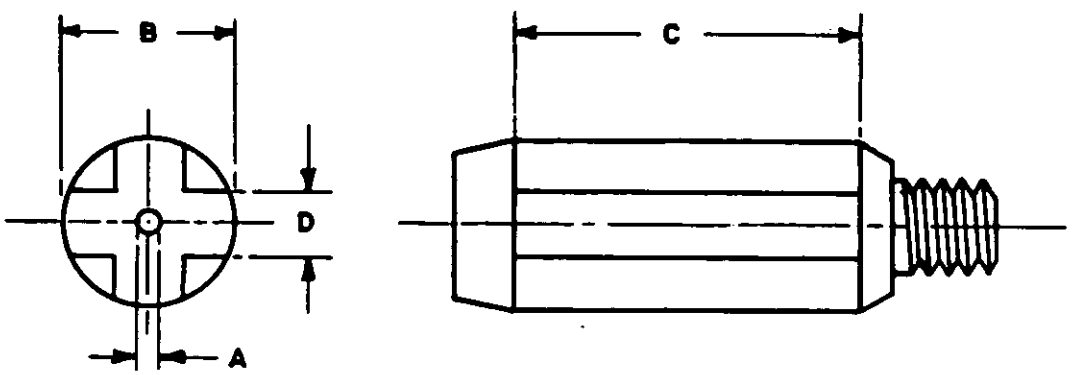
Custodians:
Army—GL
Navy—SH
Air Force—69

Review activities:
Navy—SH
Air Force—69

User activities:
Army—GL, MO, CE
Navy—YD

Preparing activity:
Navy—SH
(Project 5130-0066)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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